

Numerical study of heat transfer by laminar and turbulent natural convection in tall cavities of façade elements

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Received 28 July 2004; received in revised form 7 October 2004; accepted 1 November 2004

Abstract

Laminar and turbulent natural convection flow in a two-dimensional tall rectangular cavity heated from the vertical side has been investigated numerically for aspect ratios of 20, 40 and 80. The finite volume method was used to solve the conservation equations of mass, momentum and energy for Rayleigh numbers from 10^2 to 10^8 , the flow was considered either laminar or turbulent. For turbulent flow, four different turbulence models $\kappa-\varepsilon$ were compared along with their experimental results for a cavity with an aspect ratio of 30, it was found that the better approach was with the one reported by Ince and Launder turbulent model [N. Ince, B. Launder, On the computation of buoyancy-driven turbulent flows in rectangular enclosures, *Int. J. Heat Fluid Flow* 10 (1989) 110–117]. The average Nusselt numbers as a function of Rayleigh numbers for the aspect ratios range of 20–80 were calculated and compared with five convective Nusselt number correlations reported from the literature. Convective Nusselt number correlations for laminar flow in the range of $10^2 \leq Ra \leq 10^6$ and for turbulent flow in the range of $10^4 \leq Ra \leq 10^8$ were presented. This study will help to have more accurate heat transfer parameters for applications such as façade elements, insulating units, double-skin façades, etc.

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Keywords: Tall Cavity; Natural convection; $\kappa-\varepsilon$ model; MFVF

1. Introduction

Natural convection in cavities has been broadly studied due to many applications in engineering, such as windows with double glass, solar collectors, conservation of energy in buildings, cooling electronic devices. All of them show the importance of the processes of heat transfer. Presently, the literature review mentions that the energy consumption reduction for heating and cooling loads in buildings is an extremely important task. Thus, theoretical and experimental studies are financially supported in many countries of Europe for passive solar heating and cooling of buildings.

Passive cooling plays an important role in providing a thermally suitable environment for human comfort by natural ventilation. The multi-functional ventilated façades (MFVF) are applied to bioclimatic building design. The MFVFs are passive systems formed by an assembly of modules with two panes of different materials (opaque or semitransparent) separated by an air channel that is used to collect or evacuate solar radiation absorbed by the façade. The MFVFs are useful because the heat evacuated by the channel not only reduces energy consumption, but also decreases the temperature of façade indoors walls. Airflow in the channel can be due to natural convection or forced convection, using indoor or outdoor air. Recently, technical solutions on the MFVF are widely studied and applied to bioclimatic building design. Soria et al. [1] designed MFVF, Todorovic and Cvjetkovic [2] studied the double building envelopes and Gratia and De Herde [3] showed the optimal operation of a south double-skin façade. There are different

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theoretical models to study the thermal energy performance of ventilated façades; these models are based on simple overall energy balances [4–6]. However, a more detailed model has been reported by Manz in 2003 and 2004 [7,8]. Manz in 2003 showed numerical results of pure natural convection heat transfer of façade elements modelled as tall rectangular cavities with aspect ratios (A) of 20, 40 and 80, and Rayleigh numbers (Ra_L) from 10^3 to 10^6 . He used a commercial computational fluid dynamic (CFD) code called FLOVENT. He compared the overall heat transfer correlations calculated with five empirical ones reported from the literature. The results indicated that the overall heat transfer correlations (Nusselt numbers) obtained with FLOVENT for laminar model for 10^3 and 10^4 and turbulence model 10^5 and 10^6 agree within 20% for the reported correlations except for the correlation of Yin et al. [9]. Manz in 2004 also uses the FLOVENT code that includes convection, conduction and radiation to model double facades made of glass layers with ventilated mid pane shading device of an aspect ratio of 12. Simulation results were compared with experimental results of an outdoor test facility. The influence of layer sequence and ventilation properties are discussed and show that for a given set of layers, total solar energy transmittance vary by a factor greater than 5 [8]. However, as it is shown in the literature review, detail modelling of fluid flow and heat transfer in MFVF is complex and need more extensive study. As a first approach, we consider a detail study of fluid flow and heat transfer in a tall cavity in order to have a better approach to the empirical correlations reported in the literature for building elements. Thus, in this paper, we present a detailed bi-dimensional steady state theoretical study of fluid flow and heat transfer by natural convection in a tall cavity using a laminar model for Rayleigh numbers from 10^2 to 10^6 and turbulent model for Rayleigh numbers from 10^4 to 10^8 . Four κ – ε turbulence models were implemented to calculate the fluid flow and heat transfer in a cavity of aspect ratio of 30 and Rayleigh of 2.43×10^{10} (based on the height of the cavity) and each of their results were compared with experimental results reported by Dafa'alla and Betts [10] for the validation of the models. From this comparison, we select the best κ – ε turbulence model to represent the natural convection in tall cavities. An aspect ratio of 20, 40 and 80 were chosen for the heat transfer study in tall cavities for the Rayleigh numbers mentioned above. The results of this study are compared with the reported heat transfer correlations up to Rayleigh numbers of 5×10^6 derived from experimental data. The momentum and energy equations are solved separately for laminar and turbulent flow considering natural convection between walls using the finite volume technique.

2. Physical and mathematical model

The geometry of the two-dimensional tall cavity of width L and height H is shown in the schematic diagram of Fig. 1.

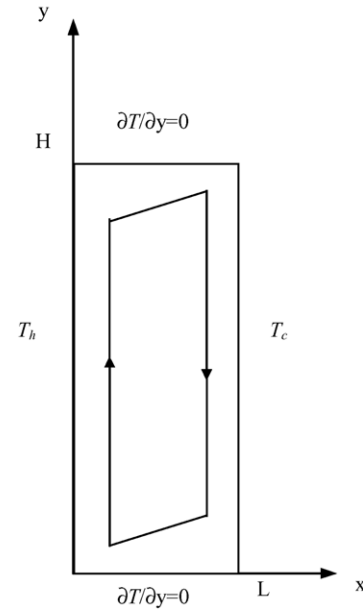


Fig. 1. Tall cavity.

The airflow inside the cavity is considered steady state laminar and turbulent. The air layer enclosed within the rectangular cavity is heated and cooled with isothermal hot (T_h) and cold (T_c) vertical walls. Two horizontal walls are insulated (zero heat flux boundary conditions at horizontal surfaces). The air properties were assumed to be constant and evaluated at a reference temperature, $T_0 = (T_c + T_h)/2$, except for the density, which is treated with the Boussinesq approximation that is considered to be valid for temperature differences up to 50 °C for buoyancy.

For two-dimensional incompressible steady state flows, the mass momentum and energy conservation equations for viscous medium are:

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (1a)$$

$$\frac{\partial(\rho u_j u_i)}{\partial x_j} = -\frac{\partial P}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} - \rho \beta (T - T_0) g_i \quad (1b)$$

$$\frac{\partial(\rho u_i T)}{\partial x_i} = -\frac{1}{C_p} \frac{\partial q_i}{\partial x_i} \quad (1c)$$

where

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \rho u'_i u'_j \quad q_i = -\lambda \frac{\partial T}{\partial x_i} + C_p \rho u'_i T'$$

and, x_i is the Cartesian coordinate system in the i -direction ($x_1 = x$, $x_2 = y$); u_i , P , T , are the velocity component, the dynamic pressure and the temperature for the laminar model, for the turbulent model u_i , P , T , are the mean velocity in the i -direction ($u_1 = u$, $u_2 = v$); the mean dynamic pressure and the mean temperature; g_i is the gravitational acceleration in the i -direction ($g_1 = 0$, $g_2 = -g$); and ρ , μ , β , λ , C_p are the density, the dynamic viscosity, the coefficient of thermal expansion, the thermal conductivity and the

specific heat at constant pressure respectively. For the turbulent model, the turbulent fluctuating velocity in the x_i -direction and the fluctuating temperature are indicated by u'_i and T' . The turbulence is modelled using the κ - ε model family because it gives a good balance between accuracy, generality and computational cost.

The turbulent stresses and the turbulent heat fluxes are written in the form:

$$\rho \overline{u'_i u'_j} = -\mu_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \frac{2}{3} \rho \kappa \delta_{ij} \quad (2a)$$

$$\rho \overline{u'_i T'} = -\frac{\mu_t}{\sigma_T} \frac{\partial T}{\partial x_i} \quad (2b)$$

where μ_t and σ_T are the turbulent viscosity and the turbulent Prandtl number respectively, and δ_{ij} is the Kronecker delta. The turbulent viscosity is related to the turbulent kinetic energy (κ) and the dissipation of turbulent kinetic energy (ε) is expressed by means of the empirical expression of Kolmogorov–Prandtl. Thus, the turbulent kinetic energy and the dissipation of turbulent kinetic energy are given from their transport equations and the resulting κ - ε equations with the Kolmogorov–Prandtl expression, after taking low-Reynolds-number effects into account, can be written as:

$$\mu_t = C_\mu f_\mu \frac{\rho \kappa^2}{\tilde{\varepsilon}} \quad (3a)$$

$$\frac{\partial(\rho \overline{u'_i \kappa})}{\partial x_i} = \frac{\partial}{\partial x_i} \left[\left(\mu + \frac{\mu_t}{\sigma_\kappa} \right) \frac{\partial \kappa}{\partial x_i} \right] + P_\kappa + G_\kappa - (\rho \tilde{\varepsilon} + D) \quad (3b)$$

$$\begin{aligned} \frac{\partial(\rho \overline{u'_i \tilde{\varepsilon}})}{\partial x_i} = & \frac{\partial}{\partial x_i} \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \frac{\partial \tilde{\varepsilon}}{\partial x_i} \right] + C_{1\varepsilon} [f_1 P_\kappa + C_{3\varepsilon} G_\kappa] \frac{\tilde{\varepsilon}}{\kappa} \\ & + E - C_{2\varepsilon} f_2 \frac{\rho \tilde{\varepsilon}^2}{\kappa} \end{aligned} \quad (3c)$$

where the variable $\tilde{\varepsilon}$, defined as $\tilde{\varepsilon} = \varepsilon - D/\rho$, is added in some turbulence models for computational convenience in order to obtain a zero value of $\tilde{\varepsilon}$ at the wall. The shear production/destruction of turbulent kinetic energy are respectively, $P_\kappa = -\rho \overline{u'_i u'_j} \partial u_i / \partial x_j$ and $G_\kappa = -\beta \rho \overline{u'_i T'} g_i$.

The following turbulence models have been employed: Jones and Launder (JL) [11]; Chien (CH) [12]; Ince and Launder (IL) [13] and Henkes and Hoogendoorn (HH) [14]. They will be referred by the acronym given in brackets. Differences between them arise in the empirical functions (f_μ, f_1, f_2), the extra terms (D, E) are the empirical constants. All of these turbulence models specify $\kappa = 0$ at the wall as boundary conditions; the ε equation specify $\tilde{\varepsilon} = 0$ at the wall except the HH model, that uses the Dirichlet boundary condition $\tilde{\varepsilon} = \infty$ (a high value).

The boundary conditions at the solid walls for the fluid velocities are zero; temperatures are specified at the vertical walls ($T = T_h$ for $x = 0$ and $T = T_c$ for $x = L$, with $T_c < T_h$); adiabatic conditions are given at the top and bottom walls ($\partial T / \partial y = 0$ for $y = 0$ and $y = H$).

From the engineering point of view, the most important characteristic of the flow is the rate of heat transfer across the cavity; this is the average Nusselt number (Nu). The Nusselt

number on the walls is calculated as $Nu = qL/\lambda\Delta T$, where $q = -\lambda(\partial T/\partial n)_{\text{wall}}$.

3. Numerical procedure

The numerical procedure used to solve the governing equations for the present work is based on the finite volume technique suggested by Patankar [15]. The flow field is discretized into cells forming a staggered grid arrangement. The general equation from which all the governing equations can be extracted is:

$$\frac{\partial}{\partial x_j} (\rho u_j \phi) = \frac{\partial}{\partial x_j} \left(\Gamma \frac{\partial \phi}{\partial x_j} \right) + S_\phi \quad (4)$$

When integrated over a finite control volume, the above general equation is converted into an algebraic equation with the following form:

$$a_P \phi_P^{n+1} = \sum_{nb} a_{nb} \phi_{nb}^{n+1} + s_\phi \Delta V + \rho^n \Delta V \phi_P^n \quad (5)$$

where n and nb denote the iteration number, and the coefficient for the neighbor grids, respectively.

Convection terms are formulated by a hybrid scheme and diffusion terms by a central scheme. The coupling between the governing equations is made by means of the SIMPLEC algorithm proposed by Van Doormal and Raithby [16]. After the finite volume approximation is used, an algebraic system of equations was obtained and solved using the line by line method (LBL). Under-relaxation is introduced by means of pseudo-transient for allowing and/or improving the rate of convergence. Global convergence is achieved when the mass balance is verified in all control volumes within a prescribed value (typically 10^{-10}) and when the residual values of the different equations are sufficiently low (typically 10^{-10}). This convergence criterion is done in order to assure good convergent solutions.

4. Validation and verification

For the purpose of validation and verification, the problem of turbulent pure natural convection of air in a differentially tall cavity with an aspect ratio of 30 has been solved for a Rayleigh number (based on the height) value of 2.43×10^{10} . This problem has been reported previously by Pérez-Segarra et al. [17], and experimental results have been reported by Dafa'alla and Betts [10]. Table 1 presents a comparison between different κ - ε turbulence models for a non-uniform grid of 81×81 and the experimental results of Dafa'alla and Betts. The results shows that the mean Nusselt number (Nu) is over predicted for all turbulence models tested (specially for the HH model, that present a difference in percentage of 90.4%), the better agreement is obtained for the IL model. The maximum vertical velocity (v_{max}^*) is generally under predicted, the better agreement is obtained

Table 1

Comparison of present solution obtained with different turbulence models with experimental data

	Turbulent model	Nu	$v_{\max}^*$ ($y = H/2$)	$\mu_{t\max}^*$
Numerical	JL	191.5 (28.5%)	0.08408 (12.1%)	29.3 (3.6%)
	IL	164.5 (10.4%)	0.09833 (2.8%)	27.9 (8.2%)
	CH	174.3 (17.0%)	0.09535 (0.3%)	37.6 (23.7%)
	HH	283.7 (90.4%)	0.07499 (21.6%)	33.5 (10.2%)
Experimental	[10]	149.0	0.09567	30.4

Note: The difference in percentage respect to the experimental data is indicated in brackets.

with the IL and CH models. The maximum turbulent viscosity ($\mu_{t\max}^*$) is well predicted for almost all the models except for the CH model. In conclusion, comparing with the experimental values, the most accurate predictions have been obtained with IL model.

Table 2 presents the comparison between the results of different models of the κ - ε turbulence models calculated and the reported ones by Pérez-Segarra et al. [17]. This table shows the values and the percentage differences of the calculated average Nusselt number (Nu), maximum Nusselt number in the hot wall ($Nu_{\max}$), maximum turbulent viscosity ($\mu_{t\max}^*$), mid-height values of vertical velocity ($v_{\max}^*$), mid-width values of horizontal velocity ($u_{\max}^*$) and the streamline at centre of cavity (ψ^*) compared with the numerical results of Pérez-Segarra et al. [17] for a non-uniform grid of 45×45 . The results indicated that the present predictions were closer to the reported ones. The maximum difference was for the CH model, for the average Nusselt number (3.4%) and maximum turbulent viscosity (2.7%). For the maximum vertical velocity, the maximum difference (1.1%) was obtained for the HH model. Being 3.4% the highest difference, we can say that our results are in agreement with the ones Pérez-Segarra et al. [17].

Table 2

Comparison of numerical results obtained with solution of Pérez-Segarra et al. [17]

	Turbulent model	Nu	$Nu_{\max}$ ($x = 0$)	$u_{\max}^*$ ($x = L/2$)	$v_{\max}^*$ ($y = H/2$)	ψ^* (at centre)	$\mu_{t\max}^*$
[17]	JL	196.1	—	—	0.08416	—	29.6
Present		198.8 (1.4%)	575.4	0.0353	0.08375 (0.5%)	−0.000790	29.7 (0.3%)
[17]	IL	166.0	570.1	0.0388	0.09836	−0.000843	28.9
Present		167.0 (0.6%)	575.4 (0.9%)	0.0386 (0.5%)	0.09846 (0.1%)	−0.000849 (0.7%)	28.3 (2.1%)
[17]	CH	180.5	—	—	0.09421	—	37.2
Present		174.4 (3.4%)	569.8	0.0391	0.09469 (0.5%)	−0.000776	38.2 (2.7%)
[17]	HH	283.1	545.4	0.0323	0.07843	−0.000775	33.6
Present		278.6 (1.6%)	542.4 (0.6%)	0.0322 (0.3%)	0.07755 (1.1%)	−0.000764 (1.4%)	33.6 (0.0%)

Note: The difference in percentage respect to the numerical results of [17] are indicated in brackets.

5. Results and discussion

After validating the numerical results with the experimental ones and verifying the numerical ones with the reported literature, a wide range of relevant parameters such as the Rayleigh numbers (based on the width) and the aspect ratio of the enclosure are analyzed in this study. Aspect ratios of 20, 40 and 80 and Rayleigh number between 10^2 and 10^8 were chosen. The temperature average between the hot and cold walls was 20°C , thus, the temperature of hot wall was 25°C and cold wall was 15°C . In this study, air is the only fluid used. The simulations were performed for both, laminar model for Rayleigh numbers from 10^2 to 10^6 and turbulent model for Rayleigh numbers from 10^4 to 10^8 .

The average Nusselt numbers as a function of Rayleigh numbers are compared with five correlations from literature, which are based mainly on experimental data. A summary of these correlations was reported in [7], those are proposed by Yin et al. [9], ElSherbiny et al. [18], Wright [19], Zhao et al. [20] and European Standard EN 673 [21].

5.1. Temperature distribution

Fig. 2 shows the temperature distribution, T^* ($T - T_c / \Delta T$), as a function of the distance, X^* (x/H), from the hot wall to the cold wall of the cavity for the aspect ratio of 20. The dimensionless temperature profile for a given vertical position ($Y^* = y/H$) are plotted for $Y^* = 0.0065$, 0.0374, 0.05, 0.9681 and 0.9935.

For the case of the Rayleigh numbers of 10^2 and 10^4 , linear dimensionless temperature distributions across the major-portion of the air layer is presented except at both ends, these linear distributions indicate that the heat from the hot wall to the cold one is transported mainly by conduction through the central core of the cavity, and by convection at the bottom of the hot wall and the top of the cold wall. On the basis of experimental results [9], the flow field corresponding with this type of temperature distribution is defined as in the “regime of conduction” as long as the horizontal temperature gradient ($\partial T / \partial x$) at the cavity centre remains at a value of about -1 .

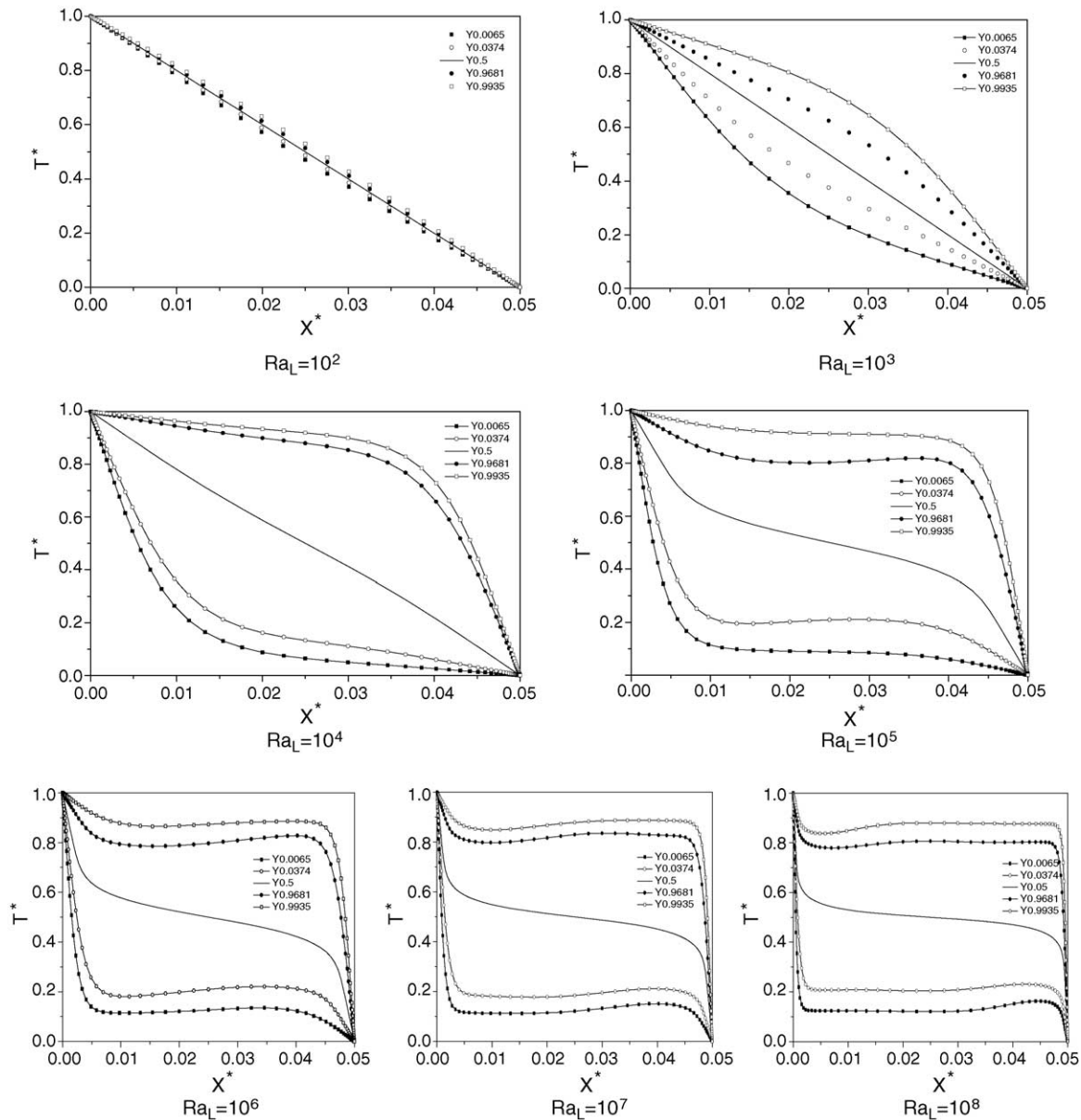


Fig. 2. Temperature profiles for aspect ratio of 20 and different positions along of vertical surface: $Y^* = 0.0065, 0.0374, 0.05, 0.9681$ and 0.9935 .

For Rayleigh numbers of 10^5 and 10^6 , the temperature profiles disclose a steep drop of temperature along the hot and cold walls with a slightly linear temperature drop in the central core. This result indicates that in addition to the heat transported by natural convection along the surface of both hot and cold walls, there is heat conduction through the central core of the layer. The flow field regime will be called “transition regime” in which the temperature gradient ($\partial T / \partial x$) at the cavity center is in the range between -1 and 0 .

For Rayleigh numbers of 10^7 and 10^8 , a steep drop of temperature in the region immediately adjacent to both hot and cold walls is observed and an almost horizontal temperature central line is presented in the central core of the cavity. This result reveals that the heat is mainly transported by natural convection from the hot to the cold wall. In this

case, the flow field under such temperature distribution at the hot and cold walls with uniform temperature at the central core of the cavity shows a “boundary layer regime”. In other words, this fluid flow boundary layer regime is characterized by central core temperature gradients at the cavity closed or equal to zero. The boundary layer regimes have strong effects in cavities with aspect ratios less than 10 and Rayleigh numbers more than 10^5 [9].

Fig. 3 shows the isotherms for different Rayleigh numbers from 10^2 to 10^8 for an aspect ratio of 80. The sequence of figures shows that the temperature distribution changes from the conduction regime to the transition regime and to the boundary layer regime. In this figure, we note that at the top and the bottom region for $10^7 < Ra_L \leq 10^8$, the boundary layer is not fully developed. However, at the

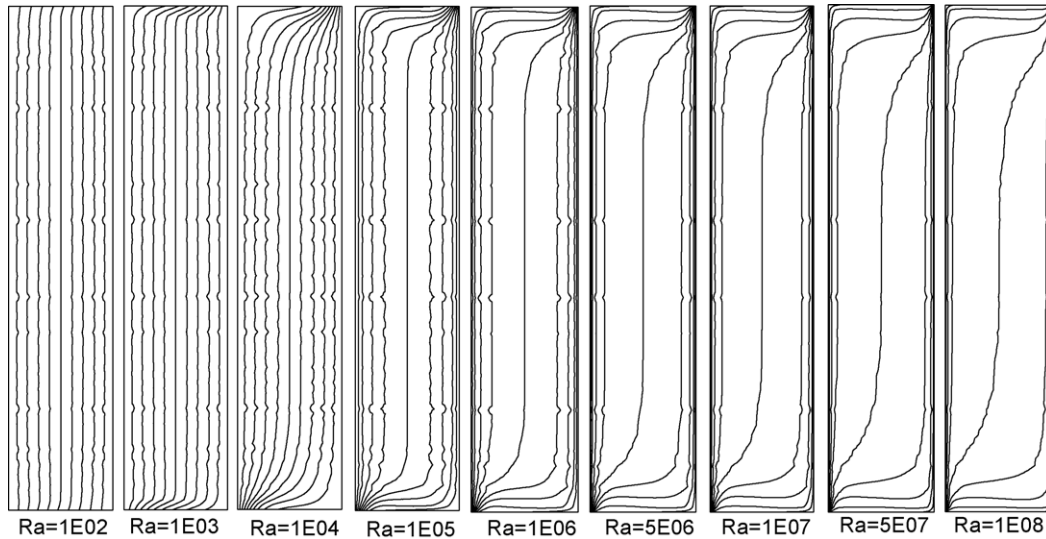


Fig. 3. Isotherms in cavities with aspect ratio of 80. Temperature distribution from conduction regime ($10^2 \leq Ra \leq 10^5$) to transition regime ($10^5 \leq Ra \leq 10^7$) and boundary layer regime ($Ra > 10^7$).

central core of the cavity the results present a significantly fully developed one-dimensional flow. In the core region the stratification is nearly zero. The turbulence in this case reaches its maximum at the vertical centerline of the cavity.

5.2. Heat transfer results

The heat transfer results are analyzed in terms of the following independent parameters: Rayleigh number (Ra_L) and aspect ratio ($A = H/L$). When considering natural convection heat transfer within cavities, the literature, in general, states that the heat transfer can be determined in functional notation as:

$$Nu = f(Ra_L, Pr, A) \quad (6)$$

In this work, the air was the only fluid used, and as the Prandtl number (Pr) does not vary significantly within the range of temperatures considered, then the Prandtl number is considered constant. Thus, the functional relationship is reduced as:

$$Nu = f(Ra_L, A) \quad (7)$$

The Fig. 4a–c show a comparison between the calculated convective Nusselt numbers and five correlations of the convective Nusselt numbers reported in the literature for aspect ratios of 20, 40 and 80 respectively. The computed results are for the laminar model in the range of $10^2 \leq Ra_L \leq 10^6$ and for the turbulent model κ – ϵ in the range of $10^4 \leq Ra_L \leq 10^8$. For aspect ratios of 20, 40 and 80, we observed that the convective Nusselt number of laminar flow model fitted very closely to the experimental results reported by Yin et al. [9] in almost all the range, contrasting to the results obtained by Manz [7], where the Nusselt numbers reported were very far from the experimental results of Yin et al. [9] in the range of $10^3 \leq Ra_L \leq 10^5$. On

the other hand, the Nu number for the turbulent model κ – ϵ fitted very closely to the results reported from Elsherbiny et al. [18] and Zhao et al. [20] in a range of $10^4 \leq Ra_L \leq 10^6$, while in the case of the comparison with EN 673 [21], the turbulent convective Nusselt number calculated was nearer in the range of $10^5 \leq Ra_L \leq 10^6$ for an aspect ratio of 20, but for aspect ratios of 40 and 80 the Nusselt number was closer in the range of $10^4 \leq Ra_L \leq 10^6$.

In Fig. 5, the average Nusselt numbers as a function of the Rayleigh numbers for different aspect ratios for (a) laminar flow and (b) turbulent flow are shown. It is seen that, the turbulent convective Nusselt number increases with aspect ratio (Fig. 5b). On the contrary, the variation of the convective Nusselt number for laminar flow decreases as the aspect ratio increases (Fig. 5a). Also, as the aspect ratio increases the Nusselt number differences decreases; for Rayleigh number of 1×10^6 , the Nusselt number percentage difference was 3.4% for laminar model and for Rayleigh number of 1×10^8 , the Nusselt number percentage difference was 1.9% for turbulent model. Velusamy et al. [22] reported that the convective Nusselt number exhibits three kinds of regimes: (i) low-growth regime up to critical aspect ratio, (ii) accelerated growth regime between critical and saturation aspect ratios, and (iii) invariant regime beyond the saturation aspect ratio. Our results fall in the regime (ii) according to Velusamy et al. [22].

From the Nusselt number calculations, heat transfer correlations are determined over the investigated laminar and turbulent models for the range of aspect ratios as:

Laminar flow ($10^3 \leq Ra_L \leq 10^6$):

$$A = 20 \quad Nu = 0.1731 Ra_L^{0.2617} \quad (5.6\%)* \quad (8)$$

$$A = 40 \quad Nu = 0.1865 Ra_L^{0.245} \quad (8.5\%)* \quad (9)$$

$$A = 80 \quad Nu = 0.1897 Ra_L^{0.2398} \quad (9.4\%)* \quad (10)$$

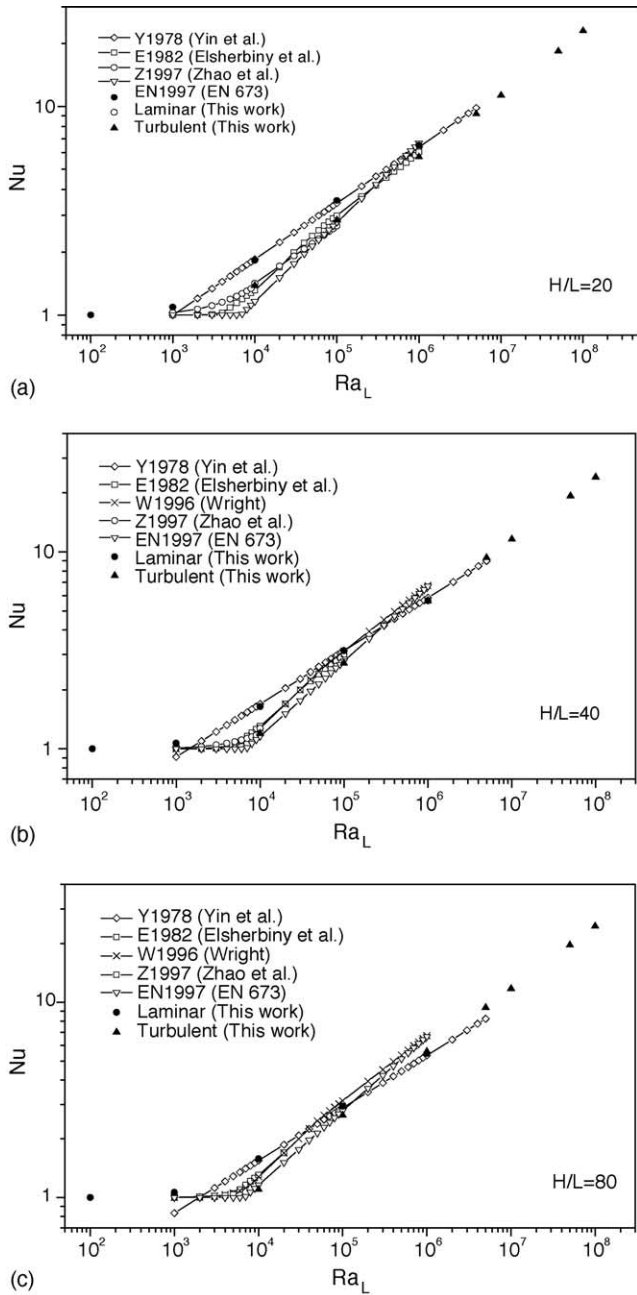


Fig. 4. Average Nusselt numbers calculated at the hot wall as a function of Rayleigh number for: (a) $A = 20$, (b) $A = 40$ and (c) $A = 80$.

Turbulent flow ($10^4 \leq Ra_L \leq 10^8$):

$$A = 20 \quad Nu = 0.0857 Ra_L^{0.3033} \quad (1.4\%)* \quad (11)$$

$$A = 40 \quad Nu = 0.0635 Ra_L^{0.323} \quad (4.0\%)* \quad (12)$$

$$A = 80 \quad Nu = 0.054 Ra_L^{0.3335} \quad (6.1\%)* \quad (13)$$

*Maximum deviations in percentage of the values predicted by the correlations given with respect to the calculated results.

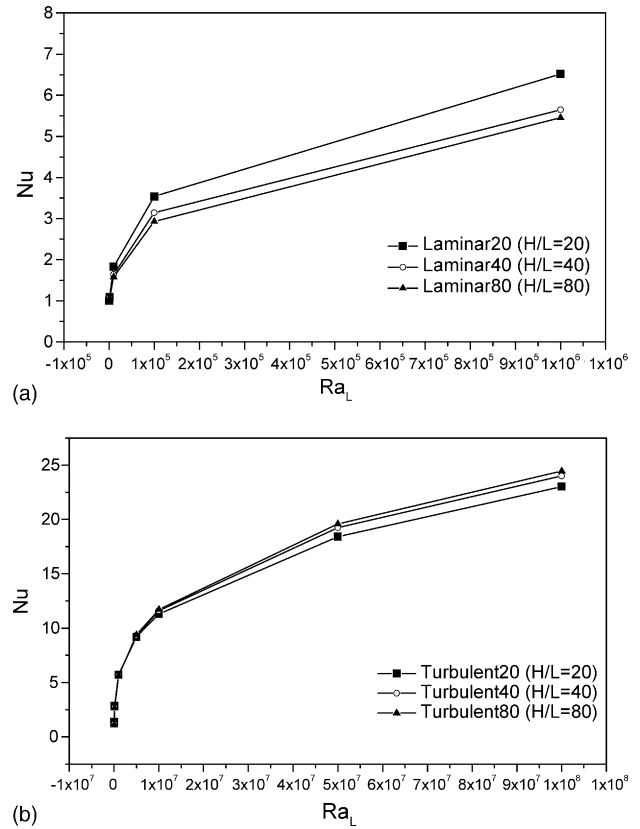


Fig. 5. Variation of Nusselt numbers with Rayleigh number for $A = 20$, 40 and 80: (a) laminar model (above) and (b) turbulent model (below).

6. Conclusions

A two-dimensional steady state numerical study has been carried out to examine the fluid flow and heat transfer by natural convection in a tall cavity using laminar and turbulent $\kappa-\epsilon$ models. Four $\kappa-\epsilon$ turbulence models were tested and compared with reported results and the IL model was selected because it predicted more accurately the experimental results.

The overall convective Nusselt numbers for the tall cavity of aspect ratios of 20, 40 and 80 were compared with five correlations based mainly on experimental results in the range of $10^3 \leq Ra_L \leq 10^6$. From this comparison, it was found that our convective Nusselt number results were very closed to ones given by the reported Nusselt number correlations. The convective Nusselt numbers for the laminar flow were in very good agreement with the ones reported by Yin et al. [9] and the convective Nusselt numbers for the turbulent flow were in good agreement with the ones of the other Nusselt number correlations reported except for the EN 673 [21] for Rayleigh number of 1×10^4 for an aspect ratio of 20.

For $A = 80$, the flow becomes almost parallel (i.e. one-dimensional) in the center region of the cavity. In the side region, two-dimensional flow was restricted near to the horizontal walls. The maximum turbulence at the vertical

centerline of the cavity. Also, it is found that the turbulent convective Nusselt number increases if the aspect ratio increases, meanwhile the convective Nusselt number for the laminar case decreases as the aspect ratio increases.

Convective Nusselt number correlations for laminar flow ($10^3 \leq Ra_L \leq 10^6$) and turbulent flow ($10^4 \leq Ra \leq 10^8$) were presented for aspect ratios of 20, 40 and 80.

Acknowledgements

The authors wish to thank Professors A. Oliva, C.D. Pérez-Segarra, K. Claramunt and J. Jaramillo at Universitat Politècnica de Catalunya for their valuable comments about turbulence models and Studentship support from CONACYT and SEP from Mexico.

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